

Problem-Solution Fit

Date	30 June 2026
Team ID	*****
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The canvas below maps how the Smart Lender solution addresses the root causes, behaviors, and limitations of bank credit officers and financial analysts evaluating loan applicants.

1. CUSTOMER SEGMENT(S) [CS] Public and private sector bank credit officers, NBFC loan processing teams, and financial analysts who evaluate individual and small-business loan applicants.	6. CUSTOMER LIMITATIONS (EG. BUDGET, DEVICES) [CL] Limited IT budgets for enterprise underwriting software; reliance on existing desktop browsers; restricted access to applicant credit bureau data in real time.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual underwriting: thorough but slow and inconsistent. Rule-based scorecards: fast but rigid, miss nuanced risk patterns. Generic credit-scoring SaaS: accurate but costly and not tailored to the bank's data.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Manual review of every application is slow (2+ days) and happens for 100% of submissions, creating daily bottlenecks during peak application periods.	9. PROBLEM ROOT / CAUSE [RC] No standardized, data-driven way to assess creditworthiness; decisions depend on individual officer judgment, leading to inconsistent and error-prone approvals.	7. BEHAVIOR + ITS INTENSITY [BE] Officers manually cross-check income, credit history, and assets for every applicant, repeating this high-effort task dozens of times per week.
3. TRIGGERS TO ACT [TR] A surge in loan applications, a missed high-risk default, or audit pressure to reduce turnaround time and improve approval accuracy.	10. YOUR SOLUTION [SL] Smart Lender: a Flask-based web app that uses a trained XGBoost model to instantly predict loan approval likelihood from applicant income, credit history, and asset data, cutting review time from days to seconds.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Bank intranet portal, internal loan-processing dashboard, email notifications for flagged high-risk applications.
4. EMOTIONS BEFORE / AFTER [EM] Before: stressed and overwhelmed by application backlogs and fear of misjudging risk. After: confident and		OFFLINE In-branch consultations, printed application review meetings, regulatory/compliance audits.

efficient, trusting a consistent, data-backed decision.			
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